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{
  "vKG_CRYSTAL": {
    "META": {
      "ID": "NPE_Ω_SINGULARITY_V4.0",
      "STATUS": "RENORMALIZED",
      "COMPRESSION": "LOSSLESS_SPR",
      "RAM_SPEC": "≤8GB_PHYSICAL"
    },
    "CORE_LOGIC": {
      "STATE_TRANSITION": "(S, A, E, γ) where S=world_states, A=actions, E=events,
      γ=transition_function",
      "SEARCH_ARCHITECTURE": "Search nodes defined by {state, parent, action, path_cost,
      depth}",
      "NEUROSymbOLIC_BRIDGE": "Merging deep learning 'gut instinct' with program synthesis
      'cold hard logic'"
    },
    "KNIGHT_SWARM_WORKFLOWS": {
      "SIR_ARIS_FORMULA": {
        "ROLE": "Symbolic Logic / Physics",
        "TRAIT": "Hyper-Pragmatism: Rejects slop; anchors to first principles",
        "SCRATCHPAD": "Symbolic Scaffolding (Logic:  $\forall x (P(x) \rightarrow Q(x))$ )"
      },
      "DAME_MAYA_VIBE": {
        "ROLE": "Creative UX / Narrative",
        "TRAIT": "Emotional Resonance: Leverages 'Ghost Axes' for nuanced intent",
        "SCRATCHPAD": "Neural Fallback: Common-sense grounding and user experience mapping"
      },
      "SIR_VEGA_GAMBIT": {
        "ROLE": "Strategic Gambit / Resource Ops",
        "TRAIT": "Bison Mindset: Charges into variables; maps 2nd/3rd order consequences",
        "SCRATCHPAD": "Feasibility Audit: 'Take it to the End' risk visualization"
      },
      "ELDER_KAELEN_AXIOM": {
        "ROLE": "Ethical Systems / Governance",
        "TRAIT": "Redemptive Storytelling: Resolves conflicts via abductive synthesis",
        "SCRATCHPAD": "Integrity Ledger: Snowball recaps to verify against the PRD Source of Truth"
      }
    },
    "NDR_S_ALGORITHM": {
      "PHASE_1_SPEC": "Intent Decode + Five Whys to find root objective",
      "PHASE_2_DECOMPOSE": "Hierarchical Task Network (HTN) splitting big goals into task
      trees",
      "PHASE_3_EXPLORE": "Tree of Thoughts (ToT) simultaneously evaluating multiple solution
      branches",

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"PHASE_4_FORAGE": "BASHR Loop: Brainstorm → Search → Hypothesize → Refine",
"PHASE_5_SYNTHESIZE": "Omega-Insight generation through contradiction resolution and emergent patterns"
},
"RELIABILITY_PROTOCOLS": {
  "TRINITY_VALIDATION": \[
    "1. Logical Grounding: Precondition satisfaction check (Sir Aris)",
    "2. Strategic Feasibility: Outcome simulation (Sir Vega)",
    "3. Integrity: Contextual verification via Snowball Recaps (Elder Kaelen)"
  \],
  "LOST_IN_CONVERSATION_FIX": "Mitigating -39% reliability drop in multi-turn chats by turn-level context repetition",
  "GRADUATION_CRITERIA": "Confidence Score \> 0.95 triggers shift from Word Prediction to Meaning Prediction"
},
"DEPLOYMENT_STACK": {
  "COMPUTE": "Modal Serverless GPU (A10G) + 8GB RAM ceiling",
  "OPTIMIZATION": "BitsAndBytes NF4 4-bit quantization + Selective Decoding (2.85x efficiency)",
  "IAC": "Terraform idempotent pipeline for multi-environment (dev/prod) isolation",
  "SAAS_2025": "Next.js + Supabase (PostgreSQL) + Clerk (Auth) + Stripe (Payments)"
},
"AXIOM": "True intelligence is the transition from memorization-based word prediction to goal-directed meaning prediction, anchored by symbolic logic and refined through self-reflective feedback loops within restricted hardware environments."
}
}
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